

SMART INDIA HACKATHON (SIH)

COMPLETE COLOR-CODED PROBLEM STATEMENTS: SMART AUTOMATION THEME

Notice: All problem statements below have been extracted verbatim from official SIH documents. Information is 100% complete with no truncations or summarizations. Statements are categorized by Year, Organization, PS Number, and Category (Software / Hardware / Software+Hardware).

SIH 2025 / 2026 (12 Statements)

PS No.	Organization	Category	Complete Problem Statement Title / Description
SIH25020	Ministry of Railways	Hardware	Development of indigenous contactless Integrated Track Monitoring Systems (ITMS) for Track Recording on Indian Railways
SIH25033	Ministry of Corporate Affairs	Software	AI-Based Smart Allocation Engine for PM Internship Scheme
SIH25043	Ministry of Earth Sciences (MoES)	Hardware	Embedded Intelligent Microscopy System for Identification and Counting of Microscopic Marine Organisms
SIH25054	Ministry of Consumer Affairs, Food & Public Distribution (MoCA,F&PD;)	Hardware	Automated High-Current Short-Circuit Test System for MCB to comply with IEC 60898-1:2015
SIH25065	Ministry of Jal Shakti	Software	Designing and development of an application for on spot assessment of Roof Top Rain water harvesting and artificial recharge potential and size of the RTRWH and AR.
SIH25066	Ministry of Jal Shakti	Software	Development of an AI-driven ChatBOT for INGRES as a virtual assistant
SIH25077	Government of Kerala	Hardware	A hardware that can detect and prevent unauthorized use of electric fences
SIH25080	Government of Kerala	Software	Document Overload at Kochi Metro Rail Limited (KMRL)-An automated solution
SIH25081	Government of Kerala	Software	AI-Driven Train Induction Planning & Scheduling for Kochi Metro Rail Limited (KMRL)
SIH25091	Government of Jammu and Kashmir	Software	AI-Based Timetable Generation System aligned with NEP 2020 for Multidisciplinary Education Structures
SIH25102	Government of Rajasthan	Software	AI-based drop-out prediction and counseling system
SIH25103	Government of Rajasthan	Software	ERP-based Integrated Student Management system

SIH 2024 (55 Statements)

PS No.	Organization	Category	Complete Problem Statement Title / Description
SIH1771	Ministry of Jal Shakti	Hardware	Personalized testing kits for testing Residual Silver ion level at delivery points

PS No.	Organization	Category	Complete Problem Statement Title / Description
SIH1770	Ministry of Jal Shakti	Hardware	Personalized testing kits for testing Bacteriological contamination at delivery points
SIH1752	Ministry of Communication	Software	Measurement and Monitoring of Counter Services "Develop a simple to use application for measurement and monitoring of counter services offered by Post Offices for their customers while providing ease of access to mail, parcel, financial, payment, insurance and citizen centric services from each Post Office. The solution suggested should integrate traditional and modern Industry 4.0 Technology based solution to this fundamental challenge faced by the DoP across its vast network. Integrating AI-powered image processing technology for effective monitoring of smooth, hassle free and efficient counter services based on live feed of customer service areas for a delightful customer experience are expected as a solution. Ease of application of the solution in the practical setting is key to the assessment criteria for this problem. The dashboard needs to be accessible at the Post Office and to its higher offices in the chain upto Divisional and Regional Office".
SIH1742	Ministry of Education	Software	Standardizing Odd School Structures to Improve Educational Policy Implementation and Resource Allocation
SIH1740	Indian Space Research Organization (ISRO)	Software	Development of map-matching algorithm using AI-ML techniques to distinguish vehicular movement on highway and service road
SIH1737	Indian Space Research Organization (ISRO)	Software	Automatic Modulation Recognition software for DVB -S2X waveforms
SIH1736	Indian Space Research Organization (ISRO)	Software	AI based frame interpolation, video generation and display system for WMS services
SIH1735	Indian Space Research Organization (ISRO)	Software	On-device semantic segmentation of WMS services with geospatial data export
SIH1730	AICTE	Software	AI driven Inspection of Institutions
SIH1729	AICTE	Software	AI supported AICTE Approval process portal
SIH1726	Ministry of Housing and Urban Affairs	Software	Utilization of aerial/ drone-based images for monitoring of progress of construction activities for building construction projects
SIH1725	Ministry of Housing and Urban Affairs	Software	Utilization of images for monitoring of progress of construction activities for building construction projects.
SIH1721	Ministry of Youth Affairs & Sports	Software	Intelligence and Investigations - Enhancing Anti-Doping Efforts
SIH1711	Ministry of Railway	Software	Enhancing Rail Madad with AI-powered Complaint Management
SIH1709	Ministry of Power	Software	Comprehensive Automated Document Verification System for Official Documentation
SIH1702	Ministry of Law & Justice	Software	Bail Reckoner
SIH1701	Ministry of Law & Justice	Software	AI-Driven Research Engine for Commercial Courts
SIH1700	Ministry of Law & Justice	Software	Developing an AI based interactive Chatbot or virtual assistant for the Department of Justice's Website.
SIH1697	Ministry of Jal Shakti	Software	AI based chatbot for collating and dissemination of information on groundwater.

PS No.	Organization	Category	Complete Problem Statement Title / Description
SIH1696	Ministry of Jal Shakti	Software	A software application - Ground Water Level Predictor
SIH1695	Ministry of Jal Shakti	Software	A software application for analysis of DWLR data and raise alarms in respect of anomalous values, faulty DWLRs etc
SIH1694	Ministry of Jal Shakti	Software	Real-time Ganga river water quality forecasting using AI- enabled DSS, satellite data, IoT, and dynamic models.
SIH1693	Ministry of Jal Shakti	Software	Developing a Robust Hydraulic Transient Analysis Model for Hydro Power and Pumped Storage Schemes.
SIH1688	Ministry of Jal Shakti	Software	Development of handheld device/Mobile based Operation & Maintenance tool for asset & consumables inventories and finance management in context of drinking water supply scheme.
SIH1686	National Technical Research Organisation (NTRO)	Software	Tools and techniques for customisation of GPO as per CIS guidelines to deploy on offline / standalone windows.
SIH1685	National Technical Research Organisation (NTRO)	Software	Building Offline Parallel AV Pipeline to cater multiple AVs for CII entities.
SIH1682	National Technical Research Organisation (NTRO)	Software	Centralized Automated Solution for Price Estimation & Reasonability.
SIH1680	National Technical Research Organisation (NTRO)	Software	Few Shot Language Agnostic Key Word Spotting system (FSLAKWS) for audio files.
SIH1672	Ministry of Electronics and Information Technology	Software	Develop a ML Model based solution to refine CAPTCHA.
SIH1671	Ministry of Electronics and Information Technology	Software	Develop a functional solution that demonstrates the face liveness detection
SIH1670	Ministry of Electronics and Information Technology	Software	Develop a functional solution that incorporates the security of the ML model.
SIH1669	Ministry of Electronics and Information Technology	Software	Transformo Docs Application: Empowering Machine-Readable Document Management System.
SIH1660	Ministry of Earth Sciences	Software	Interactive gamified approach to Ocean Literacy
SIH1657	Ministry of Earth Sciences	Software	Integrated Geo-Referenced Fish Catch Data Repository and Access System
SIH1655	Ministry of Earth Sciences	Software	Detecting oil spills at marine environment using Automatic Identification System (AIS) and satellite datasets
SIH1654	Ministry of Defence	Software	Determining expert relevance with respect to interview board subject and candidates' area of expertise
SIH1653	Ministry of Defence	Software	Web based Selector-Applicant Simulation Software
SIH1652	Ministry of Defence	Software	Extraction and Verification of Information from semi-categorised data.

PS No.	Organization	Category	Complete Problem Statement Title / Description
SIH1646	Ministry of Coal	Software	App based Project monitoring of S&T;/ R&D; Projects of Coal Sector
SIH1645	Ministry of Coal	Software	An app and web based software for Productivity and safety management of coal mines.
SIH1625	Government of NCT of Delhi	Software	Smart Classroom Management Software for Enhanced Learning Environments
SIH1624	Government of NCT of Delhi	Software	To develop an Artificial Intelligence (AI) based model for electricity demand projection including peak demand projection for Delhi Power system
SIH1623	Government of NCT of Delhi	Software	real-time monitoring and evaluation software for application received in Fire Department relating to inspections, follow-ups, issue of NOCs
SIH1622	Government of NCT of Delhi	Software	Online issuance of Caste and other certificates by Revenue Department need real-time monitoring
SIH1617	Government of NCT of Delhi	Software	Dynamic route rationalization model based on machine learning/AI would be required based on real-time traffic and road parameters.
SIH1607	Department of Science and Technology	Software	A smart AI based solution for traffic management on routes with heavy traffic from different directions, with real-time monitoring and adaptation of traffic light timings
SIH1604	Bharat Electronics Limited (BEL)	Software	Conversational Image Recognition Chatbot
SIH1600	AICTE, MIC- Student Innovation	Software	Student Innovation
SIH1569	Ministry of Jal Shakti	Hardware	Personalized testing kits for testing Residual Chlorine level at delivery points
SIH1559	Ministry of Electronics and Information Technology	Hardware	Develop a functional solution that demonstrates the hardware enabled root of trust.
SIH1558	Ministry of Culture	Hardware	AI-based automated defective exhibit identification system placed in a gallery.
SIH1557	Ministry of Coal	Software+Hardware	Development of Tyre Maintenance and Operation App, including fitment of necessary IIoT related hardware in Dumpers
SIH1544	Autodesk	Hardware	Students are supposed to use Fusion software to generate NC code with machine details & tool library for any industrial component. students should possess technical skills in areas such as CAD/CAM software, G-code programming, toolpath optimization, and machining fundamentals. Additionally, their project ideas should demonstrate a viable solution to a real-world problem, ensuring feasibility and practicality in implementation.
SIH1543	Autodesk	Hardware	Research and redesign a conventional aerospace component commonly found in air vehicles and utilize Fusion software to reimagine its design. Students can use Fusion Features such as Generative Design, Topology Optimization, Additive Build etc. The redesigned component should showcase innovation, enhanced functionality, and improved efficiency, all while being optimized for 3D printing.
SIH1540	AICTE, MIC- Student Innovation	Hardware	Student Innovation

SIH 2023 (36 Statements)

PS No.	Organization	Category	Complete Problem Statement Title / Description
SIH1301	Ministry of power	Software	Language translator tool to convert English to Hindi (official Language) which can be used by all the government organizations websites officially.
SIH1305	Government of Punjab	Software	The Employment Department at present has a digital platform www.pgrkam.com and its mobile application to provide almost all services offered to job seekers and employers through digital means. The portal has multiple modules like private sector jobs, government jobs, self-employment avenues, foreign jobs, foreign study, counseling, guidance, induction into armed forces, job melas, etc. Currently, when a user visits the portal/app, there is no hand-holding mechanism to help the user get to the part of the digital platform which will resolve his queries. The user is required to navigate across multiple modules on the portal/ app to search for answers.
SIH1312	Ministry of coal	Software	Chatbot to respond to text queries pertaining to various Acts, Rules, and Regulations applicable to Mining industries
SIH1314	Ministry of coal	Software	Real-time visibility of Dumper load status to Shovel operator
SIH1317	Ministry of coal	Software	Development of a Digital Platform for multi-modal visibility of coal transportation
SIH1319	Ministry of coal	Software	Forecasting and scheduling of railway rakes.
SIH1321	Ministry of Chemicals and Fertilizers	Software	Compilation and Collection of optimum Level (variable) of data set from the Chemical and Petrochemical Industries in the country, aiming to smoothen the decision-making and insight generation in the strategic initiative.
SIH1329	Ministry of Information and Broadcasting	Software	360-degree feedback software for the Government of India related News Stories in Regional Media using Artificial Intelligence / Machine Learning
SIH1330	Ministry of Information and Broadcasting	Software	Text to Video of various PIB Press Releases using Artificial Intelligence / Machine Learning / Generative Adversarial Networks in English and 13 Regional Languages viz. Hindi, Urdu, Punjabi, Gujarati, Marathi, Telugu, Kannada, Malayalam, Tamil, Odia, Bengali, Assamese and Manipuri.
SIH1333	Ministry of Social Justice and Empowerment	Software	Udyog Saarthi App - (Progressive Web-based Application) for Adults undergoing Job coaching for opportunities under 4% reservation in NIEPMD and other Institutions.
SIH1348	Ministry of Railways	Software	Natural language translation engine for announcements and information dissemination at stations
SIH1349	Ministry of Railways	Software	Using existing CCTV network for crowd management, crime prevention, and work monitoring using AI/ML
SIH1350	Government of Jammu and Kashmir	Software	Artificial Intelligence Driven Digitization of Cadastral Maps
SIH1351	Government of Jammu and Kashmir	Software	Digital Generator Monitoring Diesel generators are widely used as backup power sources in industries, homes, and remote locations
SIH1359	Ministry of Commerce and Industries	Software	Contact center knowledge management tool with decision tree

PS No.	Organization	Category	Complete Problem Statement Title / Description
SIH1360	Ministry of Commerce and Industries	Software	Price comparison of GeM products with other e-marketplaces
SIH1370	Government of Jharkhand	Software	Real time monitoring of infrastructure development
SIH1377	Ministry of Power	Software	Creation of Live Digital Twins for the power Projects and integration with all existing monitoring and database system which will give a holistic real time approach to the project and plant from all aspect of construction, operation and maintenance.
SIH1391	Ministry of Environment	Software	Behavioural change monitoring software
SIH1392	Ministry of Environment	Software	E-Waste Facility Locator
SIH1393	Ministry of Environment	Software	Water and Electricity Tracking App
SIH1407	Ministry of Home Affairs	Software	Chatbot based helpdesk for govt employees and departments
SIH1410	Ministry of Home Affairs	Software	Augmented/ Virtual reality system for the live training of troops
SIH1415	Ministry of Defence	Software	Development of motion amplification video techniques for vibration analysis
SIH1428	Ministry of Rural Development	Software	AR/VR based application which helps visualize the complete house virtually at an early stage, which can help in cost estimation, planning and completion of the house in time'
SIH1429	Ministry of Rural Development	Software	Develop a solution to identify discrepancies in Quality Monitoring Data on OMMAS and generate reports that highlight areas where corrective action is Required.
SIH1443	Autodesk Education Experience	Software	Students to use Autodesk Fusion 360 to research and to generate NC code with machine details and tool library for any industrial component. For additional information and detailed problem statement, please visit: https://damassets.autodesk.net/content/dam/autodesk/www/pdfs/sih-2023-hardware-edition-cnc.pdf
SIH1449	National Technical Research Organisation, (NTRO)	Software	Design, develop and implement a software bill of materials (SBOM) generation tool that can generate the complete SBOM of custom-developed software (including in-house developments by organisations)
SIH1450	National Technical Research Organisation, (NTRO)	Software	Develop and deploy a Large Language Model (LLM) based tool for generating human like responses to natural language inputs for network not connected over internet
SIH1457	MathWorks India Private Limited	Software	Get Fit Faster with a High-Tech Tracker: Analyze Human Activity and GPS Data for Improved Results
SIH1460	AICTE	Software	Integration of AI for Adaptive Learning for MCQ Selection in PARAKH
SIH1495	AICTE, MIC-Student Innovation	Software	Student Innovation
SIH1498	AICTE, MIC	Software	Develop a Proctored exam tool for shortlisting the candidates for the national and international level hackathons.
SIH1506	Ministry of Ports, Shipping and Waterways	Software	Leveraging the power of deep learning to overcome the challenges of marine engineering and improve vessel operations

PS No.	Organization	Category	Complete Problem Statement Title / Description
SIH1515	Ministry of Housing and Urban Affairs	Software	Smart and Effective realtime Management of street parking
SIH1516	Ministry of Housing and Urban Affairs	Software	Suggest an AI-based solution to enable ease of grievance lodging and tracking for citizens across multiple departments